Note: Before performing the following procedures please make sure to contact the Manufacture Parker Boiler Co. for further information. Lawrence Equipment does NOT provide tune-up service or any tools required to complete these tasks.

GENERAL THERMAL LIQUID SYSTEM CHECKLIST

WARNING

Note that Thermal Liquid oils at elevated temperatures are extremely hot and can cause burns. Exercise extreme caution when working near these systems. An oil leak at temperature can ignite.

Only trained, experienced personnel should work on Thermal Liquid Systems. Always wear safety glasses when around Thermal Liquid System.

DAILY CHECKS

The following describes the daily safety checks that should be undertaken to insure proper and safe operation of the Thermal Liquid system.

ITEM 1
Upon entering the heater room or vicinity of the heater, check for any gas leaks or smell of natural gas.

ITEM 2
Check for any thermal fluid leaks in the vicinity of the heater or piping.

ITEM 3
Make sure that no combustibles are present in, near, or around the thermal fluid heater.

ITEM 4
Insure circulating pump is running.

ITEM 5
Insure system valves are open.

ITEM 6
Check condition of combustion air vents for free air flow in and out of room.

ITEM 7
Insure air inlet to blower is clean.

ITEM 8
Note and record outlet temperature of thermal fluid heater.

ITEM 9
Note and record inlet and outlet pressure to the thermal fluid heater. Approximate pressure differential should be at least 8-20 psi.

ITEM 10
Check level of fluid in expansion tank to insure that it is safe (normal is approximately ½ full).

ITEM 11
Visually check burners with safety glasses on; observe condition of flame through observation sight hole. Use a screw driver to lift cover as it will be extremely hot. Burner should appear relatively blue not orange or lazy.

ITEM 12
Observe condition of lights on boiler panel. Note that the "Boiler Control Switch" and the "Main Burner Switch" should be on and the low fire hold switch should be on auto. These switches are located on the top of the boiler control panel.

ITEM 13
Light # 1 indicates control power.

ITEM 14
Light #2 indicates level safe in the thermal fluid expansion tank.

ITEM 15
Light #3 indicates limits are safe. These limits include the heat roll out barometric damper, high temperature, and high and low gas pressure switches.

ITEM 16
With the top three lights illuminated, the boiler will run based on operating temperature control.

ITEM 17
Light #4 indicates pilot on.

ITEM 18
Light #5 indicates main burner on.

ITEM 19
Under normal operating conditions, the boiler will modulate to match the heat load at the facility.

ITEM 20
An evaluation of the condition of thermal fluid should be performed at least one (1) time per year. This involves taking a fluid sample and having it analyzed by a lab. The lab will determine if the fluid is still good for use.

If any unsafe or abnormal conditions are found, report these immediately to boiler supervisory personnel. If in doubt, shut down heater by turning off the "Boiler Controls Switch" as well as the main gas valve to the boiler. Leave pump running.

System Shutdown
Always turn off the Heater first, and then allow the pump to run for at least one (1) hour before turning the pump off. This insures that the oil will not overheat and burn in the heater tubes.

Periodically, in addition to the daily checks of the items above, the boiler should be tuned up per Parker Procedure to 1146L. This procedure should be performed at least once a year. It encompasses a full safety check of all the operating and temperature controls on the heater to ensure safe heater operation and that burner is burning properly to meet the combustion emission requirement of the SCAQMD.